

Part C

Course title: Bayes Methods
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1. Let $\theta \in \mathbb{R}^p$ be a p -component parameter vector with prior $\pi(\theta)$ and $y \in \mathbb{R}^n$ an n -component data vector with observation model $y \sim p(y|\theta)$. the parameters are positive, and satisfy an order constraint, $0 < \theta_1 < \theta_2 < \dots < \theta_p < \infty$.

(a) [13 marks]

- (i) Consider the following proposal scheme for an MCMC algorithm targeting $\pi(\theta|y)$. Let $\theta_0 = 0$. At each step choose an index $i \sim U\{1, 2, \dots, p\}$ at random and set $\theta'_j = \theta_j$ for each $j \neq i$;
- if $i < p$ set $\theta'_i \sim U(\theta_{i-1}, \theta_{i+1})$;
 - if $i = p$ set $\theta'_p = \theta_{p-1} + \delta$ with $\delta \sim \text{Exp}(\lambda)$ and $\lambda > 0$ a fixed MCMC parameter.
- Give an MCMC algorithm using this proposal scheme. Give the acceptance probabilities in as much detail as you can.

Solution: (Straightforward application similar to examples in PS and BW)
Let $X_t = \theta$. Simulate $i \sim U\{1, 2, \dots, p\}$ and apply the proposal in the Question.
If $i \neq p$,

$$\alpha(\theta'|\theta) = \min \left\{ 1, \frac{p(y|\theta')\pi(\theta')}{p(y|\theta)\pi(\theta)} \right\}.$$

If $i = p$,

$$\alpha(\theta'|\theta) = \min \left\{ 1, \frac{p(y|\theta')\pi(\theta') \exp(-\lambda(\theta_p - \theta_{p-1}))}{p(y|\theta)\pi(\theta) \exp(-\lambda(\theta'_p - \theta_{p-1}))} \right\}.$$

Simulate $U \sim U(0, 1)$. If $U \leq \alpha(\theta'|\theta)$ set $X_{t+1} = \theta'$ and otherwise set $X_{t+1} = \theta$.

- (ii) Now consider the following proposal. Draw $u_1 \sim U(1/2, 2)$ and $u_2 \sim N(0, \sigma^2)$ where $\sigma > 0$ is a fixed parameter of the MCMC. Set $\theta' = u_1\theta + u_2$, that is $\theta'_i = u_1\theta_i + u_2$, $i = 1, 2, \dots, p$. Calculate the acceptance probability $\alpha(\theta'|\theta)$ in as much detail as you can.

Solution: (Similar to 1D PS problem but 2D here) If

$$g(u) = U(u_1; [1/2, 2])N(u_2; 0, \sigma^2)$$

and

$$\psi(\theta, u) = (u_1\theta + u_2, u')$$

then u' satisfies $\psi(\theta', u') = (\theta, u)$. That means

$$u'_1(u_1\theta + u_2) + u'_2 = \theta$$

for every θ . This has the unique solution $u'_1 = 1/u_1$, $u'_2 = -u_2/u_1$. The acceptance probability is

$$\alpha(\theta'|\theta) = \min \left\{ 1, \frac{p(y|\theta')\pi(\theta')g(u')}{p(y|\theta)\pi(\theta)g(u)} \left| \frac{\partial(\theta', u')}{\partial(\theta, u)} \right| \right\}.$$

The Jacobian for

$$(\theta, (u_1, u_2)) \rightarrow (u_1\theta + u_2, (1/u_1, -u_2/u_1))$$

is

$$\begin{aligned} \left| \frac{\partial(\theta', u')}{\partial(\theta, u)} \right| &= \begin{pmatrix} u_1 I_p & 0_p & 0_p \\ \theta & -1/u_1^2 & u_2/u_1^2 \\ 1_p^T & 0 & -1/u_1 \end{pmatrix} \\ &= u_1^{p-3}. \end{aligned}$$

The acceptance probability is then

$$\alpha(\theta'|\theta) = \mathbb{I}_{\theta_1 > 0} \min \left\{ 1, \frac{p(y|\theta')\pi(\theta')N(-u_2/u_1; 0, \sigma^2)}{p(y|\theta)\pi(\theta)N(u_2; 0, \sigma^2)} u_1^{p-3} \right\}.$$

Notice we set the acceptance probability equal zero if we shift $\theta_1 \leq 0$.

- (iii) The proposal schemes in Questions 1(a)i and 1(a)ii are combined in a single MCMC algorithm which alternates between the two updates. Comment briefly on any advantage this might give over using the scheme in Question 1(a)i or Question 1(a)ii alone.

Solution: (Simple variation of Problem Sheet question) If the target is heavy tailed the update in 1(a)i will mix very slowly. For example the mean $\bar{\theta}$ will hardly move at each update. The update in 1(a)ii will move the parameter vector in and out of the tail efficiently. The mean moves to $u_1\bar{\theta} + u_2$, so we may almost double or halve it in a single update. Similarly the statistic $\min(\theta)$ (ie, θ_1) may mix much more rapidly due to the global shift u_2 . On the other hand 1(a)ii will not be irreducible. For example the function $(\theta_p - \theta_i)/(\theta_p - \theta_1)$ is invariant under the operation of the update in 1(a)ii, so 1(a)i is certainly still needed.

- (iv) State briefly how you would choose values for λ and σ , and how you would check for MCMC convergence.

Solution: (Simple variation of Problem Sheet question) In order to choose values for λ and σ I would do some pilot runs adjusting λ and σ to give as large an ESS as possible in a run of fixed length. In practice this is done by adjusting λ and σ so that the fraction of proposals actually accepted is around 0.234. This will typically be close to optimal.

To check for MCMC convergence I would make multiple runs starting from different states and check summary statistics computed on the output agreed between runs (for eg I would overlay marginal histograms of parameters from each run - these should agree to within monte carlo error). I would estimate the ACF for each parameter and check it falls off to its asymptotic variation as lag grows. I would estimate the ESS and check it is reasonably large (where what large is will depend on the problem).

- (b) [6 marks] Consider a Markov chain simulated in the following way. Suppose $\theta^{(0)} \sim \pi(\theta^{(0)})$ is a draw from the prior and for $t = 0, 1, 2, \dots$ we generate a Markov chain by simulating $y^{(t)} \sim p(y^{(t)}|\theta^{(t)})$ and then $\theta^{(t+1)} \sim \pi(\theta^{(t+1)}|y^{(t)})$.
- (i) Calculate the joint density of $\theta^{(0)}, \theta^{(1)}$ and show that they are exchangeable.

Solution: (new)

$$\begin{aligned}
p(\theta^{(0)}, \theta^{(1)}) &= \int \pi(\theta^{(0)}) p(y^{(0)} | \theta^{(0)}) \pi(\theta^{(1)} | y^{(0)}) dy^{(0)} \\
&= \int \pi(\theta^{(0)}) p(y | \theta^{(0)}) \frac{\pi(\theta^{(1)}) p(y | \theta^{(1)})}{p(y)} dy \\
&= \pi(\theta^{(0)}) \pi(\theta^{(1)}) \int p(y)^{-1} p(y | \theta^{(0)}) p(y | \theta^{(1)}) dy
\end{aligned}$$

where $p(y) = \int \pi(\theta) p(y | \theta) d\theta$ is the marginal likelihood. The expression in the last line is symmetric in $\theta^{(0)}, \theta^{(1)}$ so $p(\theta^{(1)}, \theta^{(0)}) = p(\theta^{(0)}, \theta^{(1)})$.

- (ii) Give the transition probability density $K(\theta, \theta')$ for the chain and show the chain is reversible with respect to a stationary distribution which you should give.

Solution: (new) Since $K(\theta; \theta') = p(\theta, \theta') / p(\theta)$ with $p(\theta) = \pi(\theta)$ we have

$$K(\theta; \theta') = \pi(\theta') \int p(y)^{-1} p(y | \theta) p(y^{(0)} | \theta') dy.$$

This is reversible wrt $\pi(\theta)$ since

$$\begin{aligned}
\pi(\theta) K(\theta; \theta') &= \pi(\theta') \pi(\theta) \int p(y)^{-1} p(y | \theta) p(y^{(0)} | \theta') dy \\
&= \pi(\theta') K(\theta'; \theta).
\end{aligned}$$

- (c) [6 marks] Suppose $p = 1$ now so θ is a scalar parameter. Suppose we simulate pairs (ϕ_i, θ_i) , $i = 1, 2, \dots, K$ as follows: parameter $\phi_i \sim \pi(\phi_i)$ is an independent draw from the prior; synthetic data $y'_i \sim p(y'_i | \phi_i)$ is an independent draw from the observation model; the Markov chain in Question 1(a)iii is initialised with a draw $\theta_i^{(0)} \sim \pi^{(0)}$ from an initial distribution $\pi^{(0)}$ and run for T steps, and we set $\theta_i = \theta_i^{(T)}$ equal the last state. Let $\phi = (\phi_1, \dots, \phi_K)$ and $\theta = (\theta_1, \dots, \theta_K)$ be samples generated in this way.
- (i) Suppose $P(\theta^{(T)} | \theta^{(0)}) = \pi(\theta^{(T)} | y)$ (ie, the chain has converged by T steps for all starting states $\theta^{(0)}$). Let $p(\phi, \theta)$ be the joint distribution of the random vectors ϕ and θ . Show that $p(\phi, \theta) = p(\theta, \phi)$.

Solution: (new) Pairs are independent so $p(\phi, \theta) = \prod_i p(\phi_i, \theta_i)$ so it is sufficient to show ϕ_i, θ_i exchangeable. But we have the exact setup of Question 1(b)i (ie $\phi_i \sim \pi(\phi_i)$, $y'_i \sim p(y'_i | \phi_i)$ and $\theta_i \sim \pi(\theta | y'_i)$) where we saw they are exchangeable.

- (ii) Briefly outline a test for MCMC convergence which makes use of the result in Question 1(c)i. State any further assumptions you make to frame your test.

Solution: (new) If the MCMC is converging (so $P(\theta^{(T)} | \theta^{(0)}) = \pi(\theta^{(T)} | y)$) then ϕ and θ are exchangeable. It follows that if they do not have the same distribution then the chain cannot have converged (or at least $P(\theta^{(T)} | \theta^{(0)}) \neq \pi(\theta^{(T)} | y)$, we assume there are no bugs in code so if this fails it is due to non-convergence). Any test in which the null is $\phi \sim \theta$ and the alternative is that ϕ and θ have

distinct distributions will be appropriate. A permutation test is non-parametric and natural. Students who havnt seen these could take a paired t -test. The pairs $\phi_i - \theta_i$ are mean zero under the null. This assumes ϕ_i and θ_i are normal.

2. (a) [5 marks] Let $X_1, X_2, X_3, \dots$ be an infinite exchangeable sequence of binary random variables.

(i) What is an infinite exchangeable sequence?

Solution: (BW) The sequence is exchangeable if the distributions $p(x_1, \dots, x_n) = \Pr(X_1 = x_1, \dots, X_n = x_n)$ satisfy

$$p(x_1, x_2, \dots, x_n) = p(x_{\sigma_1}, x_{\sigma_2}, \dots, x_{\sigma_n})$$

for each $n \geq 1$ and any permutation $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n)$ of the numbers $(1, 2, \dots, n)$.

(ii) Let

$$p(x_1, \dots, x_n) = \Pr(X_1 = x_1, \dots, X_n = x_n).$$

What form does de Finetti's theorem give for the joint distribution $p(x_1, \dots, x_n)$?

Solution: (BW) There exists a distribution function $F(\theta)$ on $[0, 1]$ such that

$$p(x_1, \dots, x_n) = \int_0^1 \left[\prod_{i=1}^n p(x_i | \theta) \right] dF(\theta)$$

where $p(x_i | \theta) = \theta^{x_i} (1 - \theta)^{1-x_i}$ and

$$F(\theta) = \Pr(\Theta \leq \theta) \quad \text{where} \quad \Theta = \lim_{N \rightarrow \infty} N^{-1} \sum_{i=1}^N X_i.$$

- (b) [10 marks] A Polya urn contains b black and w white balls. At step $i = 1, 2, 3, \dots$, a ball is drawn uniformly at random from the urn. A variable X_i records the color, $X_i = 1$ if the i th ball is black, and otherwise $X_i = 0$. The ball is returned to the urn along with 1 ball of same color.

(i) Write down the probability for the sequences 1000 and 0001.

Solution: (variation of BW example)

$$p(1000) = \frac{b}{b+w} \times \frac{w}{b+w+1} \times \frac{w+1}{b+w+2} \times \frac{w+2}{b+w+3}$$

and $p(0001)$ is equal since the denominator factors are the same and the factors appearing in the numerator are the same but in the order $w(w+1)(w+2)b$.

(ii) Show that $X_1, X_2, X_3, \dots$ is an infinite exchangeable sequence.

Solution: (mentioned in lectures but not proved) Suppose $\sum_{i=1}^n x_i = K$ and the K black balls appeared at steps $i_1, \dots, i_K$ and the white at steps $j_1, \dots, j_{n-K}$. Since

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i | x_{1:i-1})$$

we have

$$p(x_{i_k} = 1 | x_{1:i_k-1}) = \frac{b+k-1}{b+w+i_k-1}$$

and

$$p(x_{j_{k'}} = 0 | x_{1:j_{k'}-1}) = \frac{w + k' - 1}{b + w + j_{k'} - 1}$$

so

$$p(x_1, \dots, x_n) = \prod_{k=1}^K \frac{b + k - 1}{b + w + i_k - 1} \prod_{k'=1}^{n-K} \frac{w + k' - 1}{b + w + j_{k'} - 1}.$$

Now the sets $\{i_1, \dots, i_K\}$ and $\{j_1, \dots, j_{n-K}\}$ are a partition of $1 : n$ (each ball was black or white) so the denominator is $\prod_{i=1}^n (b + w + i - 1)$ and

$$p(x_1, \dots, x_n) = \frac{\prod_{k=1}^K (b + k - 1) \prod_{k'=1}^{n-K} (w + k' - 1)}{\prod_{i=1}^n (b + w + i - 1)}.$$

This doesn't depend on i_k or $j_{k'}$ just K the total black balls drawn, so the sequence is exchangeable.

(iii) Let $p(x_1, x_2, \dots, x_n) = \Pr(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$. Show that

$$p(x_1, \dots, x_n) = \int \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} \text{Beta}(\theta; b, w) d\theta.$$

Hint: Note that $x\Gamma(x-1) = \Gamma(x)$, and the Beta probability density is

$$\text{Beta}(\theta; u, v) = \frac{\Gamma(u+v)}{\Gamma(u)\Gamma(v)} \theta^{u-1} (1-\theta)^{v-1}.$$

Solution: (mentioned in lectures but not proved) By the hint, $\prod_{k=1}^K (b+k-1) = \Gamma(b+K)/\Gamma(b)$ etc so from the previous part question the LHS of the relation of interest is

$$p(x_1, \dots, x_n) = \frac{\Gamma(b+w)}{\Gamma(b)\Gamma(w)} \frac{\Gamma(b+K)\Gamma(w+n-K)}{\Gamma(b+w+n)}.$$

Suppose $\sum_i x_i = K$. The RHS is

$$\begin{aligned} RHS &= \int \theta^K (1-\theta)^{n-K} \frac{\Gamma(b+w)}{\Gamma(b)\Gamma(w)} \theta^{b-1} (1-\theta)^{w-1} d\theta \\ &= \frac{\Gamma(b+w)}{\Gamma(b)\Gamma(w)} \int \theta^{b+K-1} (1-\theta)^{w+n-K-1} d\theta \\ &= \frac{\Gamma(b+w)}{\Gamma(b)\Gamma(w)} \frac{\Gamma(b+K)\Gamma(w+n-K)}{\Gamma(b+w+n)}, \end{aligned}$$

which equals the LHS.

(iv) Let $0 < m < n$ be fixed positive integers. Calculate $p(x_{m+1}, \dots, x_n | x_1, \dots, x_m)$, the posterior predictive probability for the outcome $X_{m+1} = x_{m+1}, \dots, X_n = x_n$ given data $X_1 = x_1, \dots, X_m = x_m$.

Solution: (new, simple) If $\sum_{i=1}^m x_i = K$ and $\sum_{i=1}^n x_i = K'$ then we start the second phase with $b + K$ black balls and $b + m - K$ white. Using the result

from the last part, and abbreviating $p = p(x_{m+1}, \dots, x_n | x_1, \dots, x_m)$ in order to fit things on the page,

$$p = \frac{\Gamma(b+w+m)}{\Gamma(b+K)\Gamma(w+m-K)} \times \frac{\Gamma(b+K+(K'-K))\Gamma(w+m-K+(n-m)-(K'-K))}{\Gamma(b+K+w+m-K+n-m)},$$

which is

$$p(x_{m+1}, \dots, x_n | x_1, \dots, x_m) = \frac{\Gamma(b+w+m)}{\Gamma(b+K)\Gamma(w+m-K)} \frac{\Gamma(b+K')\Gamma(w+n-K')}{\Gamma(b+w+n)}.$$

- (c) [10 marks] A random distribution $G \sim DP(\alpha, H)$ on $\Omega = [0, 1]$ has a Dirichlet Process distribution with parameter α and base distribution

$$H(d\phi) = \text{Beta}(\phi; a, b)d\phi.$$

Conditional on G , the parameters $\theta_i \sim G$, $i = 1, 2, \dots, n$ are independent and identically distributed according to G . Let $\theta = (\theta_1, \dots, \theta_n)$. Data vector $x = (x_1, x_2, \dots, x_n)$ has observation model $x_i \sim \text{Bernoulli}(\theta_i)$ independently for $i = 1, 2, \dots, n$. For given θ , let $S = \{S_1, S_2, \dots, S_K\}$ be a K component partition of $\{1, 2, \dots, n\}$ and $\theta^* = (\theta_1^*, \dots, \theta_K^*)$ a parameter vector $\theta^* \in [0, 1]^K$ satisfying $\theta_i = \theta_k^*$ for each $k = 1, 2, \dots, K$ and all $i \in S_k$. Assume α and H are given.

- (i) Give a definition of a Dirichlet Process.

Solution: (bookwork) $G \sim DP(\alpha, H)$ iff $\forall$ partitions $A_1, A_2, \dots, A_r$ of Ω

$$G(A_1), \dots, G(A_r) \sim \text{Dirichlet}(\alpha H(A_1), \dots, \alpha H(A_r))$$

- (ii) Write down the joint posterior distribution $\pi(S, \theta^* | x)$ for S and θ^* given x .

Hint: the Chinese Restaurant process determines a distribution $P(S)$ over partitions $S = \{S_1, S_2, \dots, S_K\}$ of the form $P(S) \propto \alpha^K \prod_{k=1}^K \Gamma(n_k)$ where $n_k = \text{card}(S_k)$ is the number of elements in S_k .

Solution: (simple variation of PS problem) We have in general that

$$\pi(\theta^*, S | x) \propto L(\theta^*; x) \pi(\theta^* | S) P(S)$$

where $L(\theta^*; x) = \prod_k \prod_{i \in S_k} \theta_k^{*x_i} (1 - \theta_k^*)^{1-x_i}$. Let $\bar{x}_k = n_k^{-1} \sum_{i \in S_k} x_i$. We have

$$\pi(S, \theta^* | x) \propto \alpha^K \prod_{k=1}^K \Gamma(n_k) \theta_k^{*n_k \bar{x}_k} (1 - \theta_k^*)^{n_k - n_k \bar{x}_k} B(a, b) \theta_k^{*a-1} (1 - \theta_k^*)^{b-1}$$

where $B(a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}$.

- (iii) Calculate the marginal probability $P(S | x)$ for the partition S given the data x .

Solution: (new)

$$\begin{aligned}\pi(S|x) &\propto \int \left(\frac{\alpha}{B(a,b)} \right)^K \prod_{k=1}^K \Gamma(n_k) \theta_k^{*n_k \bar{x}_k} (1 - \theta_k^*)^{n_k - n_k \bar{x}_k} \theta_k^{*a-1} (1 - \theta_k^*)^{b-1} d^K \theta^* \\ &= \left(\frac{\alpha}{B(a,b)} \right)^K \prod_{k=1}^K \Gamma(n_k) \Gamma(a + n_k \bar{x}_k) \Gamma(b + n_k - n_k \bar{x}_k)\end{aligned}$$

recognising the integral as the normalising constant of a Beta density.

3. During the 19th century there were many fatalities in coal mines. The fatality rate might be expected to drop following the introduction of new safety laws over the course of the century but might rise when extra production was demanded during wars.

For $i = 1, 2, \dots, n$, let $y_i \in (0, 100)$ be the time of the i 'th fatality. Let $y = (y_1, \dots, y_n)$. It is assumed that these times are events in a Poisson point process of rate $\lambda(t)$ deaths/year which was piece-wise constant between change points. Let $\theta_0 = 0$ and $\theta_m = 100$ and for $i = 1, \dots, m-1$ let $\theta_i \in (0, 100)$ be an increasing sequence of change points at which $\lambda(t)$ jumps up or down. The rate function is

$$\lambda(t) = \sum_{i=1}^m \lambda_i \mathbb{I}_{\theta_{i-1} < t \leq \theta_i}$$

so that $\lambda(t) = \lambda_i$ for t in the time interval between θ_{i-1} and θ_i . Denote by $\theta = (\theta_1, \dots, \theta_{m-1})$ and $\lambda = (\lambda_1, \dots, \lambda_m)$ the vectors of unknown time and rate parameters. The number of intervals m is unknown. Suppose the change-points $\theta_i, i = 1, \dots, m-1$, are modeled as arrivals in a Poisson process of unknown rate ρ with units "change-point events per year".

(a) [11 marks]

- (i) Write down the prior for m and θ given ρ , and the parameter space for (m, θ, λ) .

Solution: (similar to lecture example) Let $T = 100$. The prior $\pi(m, \theta | \rho) = \pi(\theta | m) \pi(m | \rho)$ is

$$\pi(m, \theta | \rho) \propto \frac{(m-1)!}{T^{m-1}} \mathbb{I}_{0 < \theta_1 < \dots < \theta_{m-1} < T} \exp(-T\rho) \frac{(T\rho)^{m-1}}{(m-1)!}$$

since $m-1$ is $\text{Poisson}(T\rho)$, and the θ 's are uniformly distributed in $[0, T]$ given there are $m-1$ of them, so

$$\pi(m, \theta | \rho) \propto \mathbb{I}_{0 < \theta_1 < \dots < \theta_{m-1} < T} \exp(-T\rho) \rho^{m-1}.$$

The parameter space for (m, θ, λ) is $\Omega = \bigcup_{m=1}^{\infty} \{m\} \times \Lambda_m \times \Theta_{m-1}$ where $\Lambda_m = [0, \infty)^m$, $\Theta_0 = \emptyset$ and $\Theta_{m-1} = \{\theta \in [0, \infty)^{m-1} : 0 < \theta_1 < \dots < \theta_{m-1} < T\}$.

- (ii) Data for the annual fatalities 1790-1799 and 1901-1910 are available. Mining companies were conservative and the kind of significant change in mining methods which would lead to a measurable change in fatality rates cannot have been common. Explain how you would elicit a prior for λ given m and a prior for ρ , stating any assumptions you make and any further information you need to complete this task.

Solution: (application of lecture on elicitation, similar to PS question) The priors for λ and ρ should cover the positive reals. Suppose we take a $\Gamma(\alpha, \beta)$ distribution for ρ (we don't have the info to elicit anything more complex). As the question hints, the number of change points in T years must be fairly small, so a value $\rho = 1$ which gives $\mathbb{E}(m | \rho) = T$ is very large (one change point expected each year). Suppose $p_2 = \pi(m \geq T)$. There should also be a reasonable probability $p_1 = \pi(m = 0)$ for no CPE's in the 100 years. We could ask for more detail on the values of these numbers and set α, β to achieve them. We could similarly take $\Gamma(a, b)$ priors for each $\lambda_i, i = 1, \dots, m$. I would treat these as exchangeable and independent given a, b . The data for 1790-1799 and 1901-1910 give us an estimate of the mean a/b . If we take $a = 1$ we

have the maximum entropy prior with this mean. I would simulate $T' = 20$ years of synthetic data, taking $\rho \sim \Gamma(\alpha, \beta)$, $\theta \sim PPP(\rho)$ and for $i = 1, \dots, m$, $\lambda_i \sim \Gamma(1, b)$ and $y' \sim PPP(\lambda(t))$ for $0 < t < T'$ and check y' resembles the 1790-1799 and 1901-1910 data in distribution. I would focus on adjusting a in the λ prior to try to improve the match.

- (iii) Write down the posterior $\pi(\lambda, \theta, m, \rho|y)$ in terms of the available model elements. What is the parameter space for this posterior?

Solution: (new) Let $n_j = \sum_i \mathbb{I}_{\theta_{j-1} < y_i \leq \theta_j}$ be the number of events in the interval $(\theta_{j-1}, \theta_j]$. This number is $\text{Poisson}(\lambda_j(\theta_j - \theta_{j-1}))$ and the points are otherwise scattered uniformly at random. The likelihood is

$$\begin{aligned} L(\theta, \lambda; y) &= \prod_{j=1}^m e^{\lambda_j(\theta_j - \theta_{j-1})} \frac{(\lambda_j(\theta_j - \theta_{j-1}))^{n_j}}{n_j!} \times \frac{1}{(\theta_j - \theta_{j-1})^{n_j}} \\ &= \prod_{j=1}^m e^{\lambda_j(\theta_j - \theta_{j-1})} \frac{\lambda_j^{n_j}}{n_j!} \end{aligned}$$

so the posterior is

$$\pi(\lambda, \theta, m, \rho|y) \propto \pi(\rho)\pi(m, \theta|\rho)\pi(\lambda|m)L(\theta, \lambda; y)$$

with the likelihood given above, $\pi(\lambda|m) = \prod_{i=1}^m \Gamma(\lambda_i; a, b)$, $\pi(\rho) = \Gamma(\rho; \alpha, \beta)$ and $\pi(m, \theta|\rho)$ given in the solution to Question 3(a)i. The parameter space is $(\lambda, \theta, m, \rho) \in \Omega \times [0, \infty)$ with Ω defined above.

- (b) [7 marks] Suppose you have a Metropolis Hastings MCMC algorithm targeting $\pi(\lambda, \theta, m, \rho|y)$, realising samples $\lambda^{(k)}, \theta^{(k)}, m^{(k)}, \rho^{(k)}, k = 1, \dots, K$ returned from the MCMC.
- (i) Suppose the unknown true fatality rate at time t is $\lambda^*(t)$ and the unknown true number of change points is $m^* - 1$ (so $\lambda^*(t)$ is piecewise constant over m^* intervals). Suppose the loss function for estimating $\lambda^*(t)$ at $t \in (0, 100)$ with $\lambda(t)$ is $(\lambda^*(t) - \lambda(t))^2$, and the loss for estimating m^* with m is $\mathbb{I}_{m^* = m}$. State briefly how you would estimate m^* and $\lambda^*(t)$ minimising the expected posterior loss for each quantity, using the MCMC output.

Solution: (similar to PS problem) The estimator minimising the mean square error (ie the EPL for this loss) is the posterior mean. In this setting the model index m is just another parameter, so we want to estimate $\mathbb{E}_{\lambda, \theta, m, \rho|y}(\lambda(t))$. Let

$$\lambda^{(k)}(t) = \sum_{i=1}^m \lambda_i^{(k)} \mathbb{I}_{\theta_{i-1}^{(k)} < t \leq \theta_i^{(k)}}$$

so averaging over everything

$$\hat{\lambda}^*(t) = K^{-1} \sum_{k=1}^K \lambda^{(k)}(t).$$

The estimator minimising the 0-1 loss function is the posterior mode. This is estimated by the empirical mode of the simulated $m^{(k)}, k = 1, \dots, K$.

- (ii) How would you make a goodness of fit check for this problem? Give the details of any procedures or test statistics you would use.

Solution: (similar to Lecture and PS examples) One approach would be to simulate synthetic data from the posterior predictive distribution, $p(y'|y)$. This may be done by simulating data sets $y^{(k)} \sim L(\theta^{(k)}, \lambda^{(k)}; \cdot)$. We should form some test statistic $T(y)$ which we think is likely to be sensitive to model misspecification. The log-likelihood is a good catch-all but I would expect students to find it hard to express the likelihood for the real data at the posterior mean $\hat{\lambda}^*(t)$, $0 \leq t \leq T$, since this is not piece-wise constant. If we divide the interval $[0, T]$ up into 10 year bins, $b_0 = 0, b_1 = 10, \dots, b_{10} = 100$, then there is a certain number of events in each bin $B_j(y) = \sum_{i=b_{j-1}}^{b_j} \mathbb{I}_{B_{j-1} < y_i \leq B_j}$, $j = 1, \dots, 10$ where $n(y)$ is the dimension of y . Let $B(y) = (B_1(y), \dots, B_{10}(y))$. Take $T(y) = \text{mean}(B(y))/\text{var}(y)$ as our test statistic. This is a measure of (under) dispersion. Now compute $T^{(k)} = T(y^{(k)})$ and compare this with $T^* = T(y)$. We can for example report a “p-value”, that is $p = K^{-1} \sum_k \mathbb{I}_{T^{(k)} > T^*}$. Values close to 0 or 1 indicate misspecification.

One simple alternative approach would be make a histogram of the number of events in each d -years (where d is chosen to get a reasonable smooth histogram) and plot this over $d\hat{\lambda}^*(t)$ plotted as a function of t . The curve and histogram should be in reasonable agreement.

- (c) [7 marks] Now suppose the λ prior has the form $\pi_\Lambda(\lambda) = \prod_{i=1}^m \pi(\lambda_i)$ and, in all changes to λ in the MCMC algorithm targeting $\pi(\lambda, \theta, m, \rho|y)$, the update changes a single λ_i -value, with the proposal λ'_i a draw from the prior. There is an hypothesis that $\lambda(t)$ is decreasing, so that $\lambda_{i+1} < \lambda_i$ for $i = 1, \dots, m$. Let $\Lambda'_m = \{\lambda \in [0, \infty)^m : \lambda_{i+1} < \lambda_i, i = 0, \dots, m\}$ and $\Lambda = \bigcup_{m=1}^\infty \Lambda_m$.
- (i) How would you modify the MCMC algorithm to target $\pi(\lambda, \theta, m, \rho|y, \lambda \in \Lambda')$? It is not necessary to give details of revised acceptance probabilities.

Solution: (new) Initialise the MCMC with a state satisfying $\lambda \in \Lambda'$ (for example, a sorted draw from the prior). The simplest approach is simply to reject any proposal satisfying $\lambda \notin \Lambda'$. This will be inefficient if the number of intervals m is large or the steps in λ are small. Alternatively we may adjust the proposals so that $\lambda'_i \sim \pi(\lambda'_i | \lambda_{i-1} < \lambda'_i < \lambda_{i+1})$. This would not lead to any change in the acceptance probability for fixed dimension updates but the variable dimension updates would pick up factors like $\pi(\lambda_{i-1} < \lambda'_i < \lambda_{i+1})$.

- (ii) Denote by $B_{1,2}$ the Bayes factor measuring the support for model (1) $\lambda \in \Lambda'$ over model (2) $\lambda \in \Lambda$. Show that

$$B_{1,2} = \frac{P(\Lambda'|y)}{\mathbb{E}(1/m!)}$$

where $P(\Lambda'|y)$ is the probability $\lambda \in \Lambda'$ in the constrained model and $\mathbb{E}(1/m!)$ is an expectation which you should define.

Solution: (new)

$$\begin{aligned} B_{1,2} &= \frac{P(\lambda \in \Lambda'|y)P(\lambda \in \Lambda)}{P(\lambda \in \Lambda|y)P(\lambda \in \Lambda')} \\ &= \frac{P(\lambda \in \Lambda'|y)}{P(\lambda \in \Lambda')} \\ &= \frac{P(\Lambda'|y)}{\mathbb{E}(1/m!)} \end{aligned}$$

since $P(\lambda \in \Lambda')$ is the probability $\lambda_1, \dots, \lambda_m$ are in decreasing order, which is $1/m!$, and $m-1$ itself is $\text{Poisson}(\rho T)$, so $P(\lambda \in \Lambda') = \mathbb{E}(1/m!)$ which is

$$\mathbb{E}(1/m!) = \int \pi(\rho) \left[\sum_{m=1}^{\infty} \frac{1}{m!} \exp(-T\rho) \frac{(T\rho)^{m-1}}{(m-1)!} \right] d\rho.$$